

Graph Learning for Interpretable Urban Accessibility and Opportunity

Project proposal

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1 Problem definition

Cities are complex, interconnected systems where access to opportunities is shaped by spatial structure, infrastructure, and social dynamics. Graph learning offers a powerful framework to model these relationships, enabling interpretable insights into urban accessibility and the distribution of opportunity. The presence of multimodal data that integrate GTFS transit schedules, sociodemographics (ACS), employment locations (LEDH) and land-use maps (OSM) allow us to compute accessibility indices (e.g. number of jobs reachable within 30–60 min by transit or car).

The goal is to train, among others, tree-based graph learning models (GraphSAGE, GCN) to predict these accessibility scores from spatial features (e.g. population density, land-use mix, transit stop density, travel speeds) and then explain the results with the help of GN-NEExplainer and partial-dependence plots. The explainable outputs support equity-focused planning by showing which factors (e.g. lack of transit stops or mixed-use) most depress access in underserved areas.

2 Keywords

Urban accessibility, Graph neural networks (GNN), GraphSAGE, Graph convolutional networks (GCN), XGBoost, Explainable AI (XAI), SHAP values.

3 Literature review

A list of relevant literature sorted by estimated relevance:

- **A hybrid spatial learning method integrating graph neural networks and explainable AI to understand urban vitality (Xie et al., 2025)** is very similar to this project. It tries to predict vitality with graph learning, using on a list of spatial and socio-economic features that are determined per urban block, which are represented by nodes in the graph. It also extracts features from street view imagery. With this graph, it uses GraphSAGE to generate low-dimensional node embeddings,

followed by a linear layer to predict the urban vitality for each node. To explain the model it uses GNNExplainer.

- **Modeling walking accessibility to urban parks using Google Maps crowdsourcing database in the high-density urban environments of Hong Kong (Gong, 2023)** talks about a way to model the accessibility of parks in an urban environment using Google Maps crowdsourcing data (Google Maps Distance Matrix API). Its results show that just looking at park size does not give a good representation of the parks attractiveness, adding accessibility give a much better representation. This can be useful in this thesis when modelling accessibility.
- **Inductive Representation Learning on Large Graphs (Hamilton et al., 2018)** explains how GraphSAGE works.
- **GNNExplainer: Generating Explanations for Graph Neural Networks (Ying et al., 2019)** explains how GNNExplainer works.
- **Modeling job accessibility using online map data: An extended two-step floating catchment area method with multiple travel modes (Xiao et al., 2021) and Evaluation of job accessibility of Shenyang using gaussian based 2-step floating catchment area method (Qi et al., 2015)** show the use of the 2FSCA method to define job accessibility. This can be used in our research to serve as a target variable for the model to predict.
- **Graph learning-based spatial-temporal graph convolutional neural networks for traffic forecasting (Hu et al., 2021)** talks about the use of a GCN to predict traffic. It shows good results while saving training time and computational resources.

4 Research Questions

There are two research questions proposed in this thesis:

1. How accurately can graph learning models predict multimodal accessibility indices from spatial and socio-economic features across urban environments?
2. Which spatial factors most strongly influence urban accessibility, and how can interpretable methods (e.g., GNNExplainer) reveal distinct patterns of opportunity.

5 Method and Approach

To complete the two research questions I propose 4 main steps:

5.1 Data collection and preprocessing

To construct a graph-based representation of the urban environment, multiple datasets will be collected and combined into node-level features. A key design decision is the spatial unit of analysis. I propose the use of either census blocks, census tracts, or a regular grid. The final choice will depend on data availability and computational tractability. Finer units (e.g., blocks) provide more detail but may introduce sparsity and noise, whereas bigger units (e.g., tracts) improve stability at the cost of spatial precision.

5.1.1 Data sources

The analysis will use several open source datasets covering different aspects:

- **Sociodemographic data (e.g., ACS):** Variables like population density, income, employment status, gender can be used as node attributes.
- **Transport data (e.g., GTFS):** Public transit schedules, transit routes and the location of transit stops can be used to compute travel times and transit accessibility.
- **Land-use and urban form (OpenStreetMap):** Features such as land-use mix, road density/length, intersection density, and amenity distribution can be extracted.
- **Employment data (e.g., LEHD):** Spatial distribution of jobs can serve as the basis for defining opportunity. This could be done by counting the number of jobs within 1, 2, 3, 4, and 5 km from its centroid, also the average income level could be extracted.

The exact combination of features will depend on data availability and effectiveness. A design choice is whether the features will only include data from inside the spatial unit, or also include data from an area around it. Inspiration can be gathered from Xie et al., 2025.

5.1.2 Definition of accessibility

Accessibility will have to be defined, an option would be to get the number of opportunities (e.g., jobs) reachable within a given travel time, using different transport modes. Travel times can be computed using the Google Maps Distance Matrix API as done by Gong, 2023.

Another approach would be to use the two-step floating catchment area (2SFCA) method, in the same way that Xiao et al., 2021 and Qi et al., 2015 have done. This method a more balanced demand-supply relationship of job seekers and providers.

5.1.3 Feature engineering and graph construction

All datasets will have to be spatially aligned to the chosen unit of analysis (e.g., census block, census tract). This involves things like: aggregating point-based data (e.g., amenities, jobs) to spatial units, computing density-based features (e.g., population density, stop density) and deriving urban form indicators and network connectivity metrics. In addition, accessibility itself will serve as the target variable for supervised learning.

The urban area will be represented as a graph, several edge definitions will be considered, options are adjacency-based edges and distance-based edges.

5.2 Graph learning and prediction

In this step, graph learning methods are used to incorporate spatial relationships between areas and to learn richer representations of each node. Graph learning models to be considered:

- **GraphSAGE:** Aggregates information from neighboring nodes to learn node embeddings and is suitable for large graphs.

- **Graph Convolutional Network (GCN):** Capture spatial dependencies through learned convolution filter on the graph.

After that the learned embeddings are used as input for a separate model like a linear layer or a small neural network. Would also be interesting to look at effectiveness of these trained models on other cities.

5.3 Explainability

To understand which factors drive accessibility, interpretability methods can be applied to the trained models. For models such as XGBoost, SHAP values can be used to quantify the contribution of each input feature to the predictions. Since embeddings learned by graph neural networks are not directly interpretable, SHAP analysis will focus primarily on models that include the original input features. Embeddings could still be included to improve predictive performance.

To explain the graph neural networks GNNExplainer will be used. This method provides local explanations by identifying which node features and neighboring nodes are most influential for a given prediction.

5.4 Visualization

To communicate the results, an interactive visualization will be developed. This may take the form of a dashboard that displays spatial patterns of predicted accessibility, feature importance, and explanation outputs.

The visualization will include maps showing accessibility levels across the study area, as well as tools to inspect individual locations and their driving factors. Inspiration will be taken from existing systems such as <https://whatif.sony CSL.it/index.html>.

6 Evaluation

Models will be evaluated on their ability to predict accessibility indices using standard metrics such as mean absolute error (MAE), root mean squared error (RMSE), and R-squared. The dataset will be split into training and test sets, for example an 80/20 split of nodes, to assess generalization. Additionally, cross-validation and area-based holdout can be used to evaluate how well the models transfer to unseen urban areas. Another option would be to gather data over multiple years and use earlier years to predict later years.

To test the robustness of the models, trained models could also be applied to other cities not seen during training. This will provide insights into the transferability of learned relationships between spatial features, sociodemographics, and accessibility.

The XGBoost model will serve as a baseline model to evaluate the graph learning models.

7 Plan

Week	Deadlines	Activity
1 (W14)	-	Literature review & Write project proposal
2 (W15)	Project plan inleveren, DataNose project aanmaken (13 april)	Literature review & Collect datasets
3 (W16)	-	Collect datasets & Determine features and accessibility definition.
4 (W17)	-	Collect datasets & Finish city graphs
5 (W18)	-	Start writing first thesis draft
6 (W19)	First draft thesis inleveren (8 mei)	Implement prediction algorithms (GraphSAGE, GCN, XGBoost)
7 (W20)	Midterm presentaties	Implement graph learning algorithms (GraphSAGE, GCN, XGBoost) & Prepare midterm presentation
8 (W21)	Midterm presentaties	Implement explainability methods
9 (W22)	-	Implement explainability methods
10 (W23)	Go-No-version indienen (5 juni)	Create visualization of results & Write thesis
11 (W24)	-	Write thesis
12 (W25)	Final Thesis indienen, Delta-document (19 juni)	Write thesis
13 (W26)	Defenses (22 juni - 10 juli)	Prepare defense

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